Master's Thesis Proposal

Harvard SEAS / IACS

Submitted: May 9, 2026
Due: April 10, 2026

1. Title

Attention-Guided Router Steering: A Feature-Learning Lens on Behavioral Intervention in Sparse Mixture-of-Experts Language Models.

2. Advisor

Adityanarayanan (Adit) Radhakrishnan, Assistant Professor, MIT Department of Mathematics; Associate Member, Broad Institute of MIT and Harvard.
aditradha.com *Primary research supervisor.*

Note: I will identify a Harvard SEAS internal sponsor or co-advisor as required by IACS administrative policy; pending confirmation with Daniel Weinstock.

3. Abstract

Sparse mixture-of-experts (MoE) language models such as Mixtral 8x7B route each token to a small subset of feed-forward experts at every layer. Recent work (SteerMoE) has shown that contrastive prompt pairs can identify experts whose activation rate differs in the presence of a target concept, and that adding a fixed bias to those experts' router logits can shift generations behaviorally without any fine-tuning. In a faithful re-implementation we find that this intervention transfers weakly to Mixtral on a question-only concept-transfer test, and that the underlying signal — the (layer, expert) activation difference — is diffuse: the top twenty cells out of $32 \times 8 = 256$ explain less than 10% of the total $|\Delta|$ mass. This thesis proposes **Attention-Guided Router Steering (AGRS)**, a drop-in replacement for SteerMoE in which the per-token contribution to the risk-difference readout is weighted by the attention mass that token sends toward a concept-bearing prefix span. We develop AGRS, evaluate it against SteerMoE and three hidden-state activation-steering baselines (ITI, ActAdd, RepE) across four MoE checkpoints and four concept families, and use the *average gradient outer product* (AGOP) framework from my advisor's recent work on feature learning to formalize *which* features the router-bias intervention is actually moving. The deliverables are a NeurIPS-track method paper, an open-source replication artifact, and a thesis monograph that positions router steering inside a feature-learning theory of neural network behavior.

4. Motivation and Background

Personal background. I am a Harvard IACS master's student with a research interest in the mechanistic interpretability of large language models, particularly the question of how, where, and through what mechanisms behavioral interventions on frozen models actually work. I have been running empirical experiments on SteerMoE-style routing-bias interventions on Mixtral 8x7B-Instruct on the MIT ORCD cluster, and a working draft of the conference paper that emerged from those experiments is checked in alongside this proposal.[1] That draft is a small diagnostic note. The thesis will turn it into a method paper.

Why this project. Three observations drove me to this topic:

- Steering is the cleanest empirical wedge into the question of *what features a model has learned*. If we can move a generation toward a concept by adding a bias to a small set of scalar parameters at inference time, then those parameters represent that concept in a behaviorally-meaningful sense.
- Sparse MoE language models are an unusually clean substrate for that wedge: every layer's routing decisions are a discrete, inspectable structure, unlike the continuous residual stream of a dense transformer.
- Existing steering methods read off the contrastive signal using fixed token positions or unweighted averages over a span. Attention-guided readouts are the obvious next step, and have been demonstrated for hidden-state interventions but not for router-logit interventions on MoE models.

Connection to my advisor's research. Adit's recent papers — *Mechanism for Feature Learning in Neural Networks* (*Science*, 2024), *Mechanism of Feature Learning in Convolutional Neural Networks* (arXiv:2309.00570), and *Linear Recursive Feature Machines Provably Recover Low-rank Matrices* (arXiv:2401.04553) — introduce and develop the **Average Gradient Outer Product (AGOP)** as a mathematical mechanism by which neural networks acquire features. AGOP gives us a principled handle on the question this thesis turns on: *which input directions does a model treat as informative for a given prediction*. The same mechanism lets us ask *which routing decisions a Mixtral expert is sensitive to*, which is the readout question I want to answer. The thesis proposes to bring AGOP and Recursive Feature Machines (RFM) to bear on the steering-readout problem, using my advisor's existing toolkit on a problem domain (behavioral steering of MoE LLMs) where it has not yet been applied.

Future plans. I intend to pursue a PhD in machine-learning interpretability after completing this thesis. The thesis is therefore designed to produce a publication-track research artifact (a NeurIPS-tier method paper), hands-on experience with cluster-scale generation experiments on billion-parameter MoE models, and a written monograph that demonstrates the ability to formalize, implement, and rigorously evaluate a method contribution end-to-end.

5. Goals

The thesis has three concrete goals, each with a specific deliverable.

[1] See `paper/docs/paper/main.pdf` in the project repository, github.com/pgrindehollevik-harvard/MoE_attention_guided_steering.

- **Method.** Define AGRS formally as an extension of SteerMoE in which the per-token contribution to the per-(layer, expert) risk-difference table is weighted by attention mass to a concept-bearing prefix span. Implement AGRS as a drop-in replacement in the existing Mixtral and OLMoE backends, with unit tests verifying that AGRS reduces to SteerMoE when the attention weights are uniform.

- **Empirical evaluation.** A statistically powered, pre-registered comparison of AGRS against five alternatives — unsteered baseline, SteerMoE, ITI, ActAdd, and RepE difference-of-means — across four MoE checkpoints (Mixtral 8x7B-Instruct, OLMoE 1B-7B-Instruct, Qwen2-MoE 14B-A2.7B, DeepSeek-V2-MoE-Lite if licensing permits) and four concept families (fears, professions, refusal targets, sentiment polarity). Primary metric is concept-transfer rate (CTR) under an LLM-as-judge evaluation cross-validated against a human-rater subset. Statistical reporting is paired bootstrap 95% CIs and pre-registered paired t-tests on seed-collapsed deltas.

- **Mechanistic analysis.** Use AGOP and RFM-flavored diagnostics to ask *why* attention-weighted readout concentrates the risk-difference signal where uniform readout does not. Specifically, characterize the gradient structure of the selected experts with respect to the concept-bearing prefix tokens and ask whether AGOP-implied feature directions align with the experts that AGRS selects.

Expected outcomes. A submitted NeurIPS or ICLR main-track paper; an open-source replication artifact (Apptainer image, pinned model SHAs, full generation logs, judge transcripts); and a thesis monograph organized around the three goals above with a final chapter situating the empirical findings inside Adit's feature-learning framework.

6. Scientific Justification

Why this is interesting. Behavioral steering of LLMs at inference time is one of the few areas of mechanistic interpretability where small, tractable interventions on frozen models produce measurable behavioral effects. Most published steering methods operate on the residual stream of dense transformers. Sparse MoE models give us a distinctively different intervention site — the router logit — whose downstream causal effect is mediated through a discrete, inspectable selection over a small expert pool. Whether the readout-and-intervene paradigm transfers from dense to sparse architectures, and whether it does so with the same readout rules, is an open empirical question. My preliminary results suggest the answer is *not under uniform readout*, which gives this thesis an unusually crisp method-comparison story: the readout, not the intervention site, is doing the work.

Why this is innovative. Three innovations distinguish the proposed thesis from existing work:

- *Method.* AGRS is, to my knowledge, the first attention-guided readout developed specifically for router-logit interventions on sparse MoE models. The reduction from AGRS to SteerMoE (under uniform attention) makes the contribution isolatable from confounding implementation differences — the same intervention site, the same generation pipeline, the same experiment harness, the same evaluation. Only the readout changes.

- *Mechanistic framing.* I will use Adit's AGOP machinery to formalize the feature directions that selected experts respond to, then test whether AGRS's expert selections align with those directions on the same concept-prompt pairs. This is a use of the feature-learning toolkit on a behavioral-intervention problem where it has not been applied.

- *Empirical rigor.* The evaluation is pre-registered with a Phase-3 decision gate (ΔCTR $\geq +0.10$ over SteerMoE on the pilot set) before the more expensive cross-model and cross-concept phases run. The full prompt set, judge transcripts, and seed sweeps are released under CC-BY at submission.

7. Literature Review

I group the relevant literature into four threads.

(i) Activation steering on dense transformers. *Inference-Time Intervention* (Li et al., *NeurIPS* 2023) adds a learned direction to a small set of attention heads to reduce hallucination on TruthfulQA. *Representation Engineering* (Zou et al., 2023) formalizes the readout-and-edit problem on contrastive prompt pairs at the residual stream. *Activation Addition* (Turner et al., 2023) projects the contrastive signal across multiple residual layers via projected gradient descent. *Attention-Guided Steering* (2024) refines *where* on a residual stream to read by weighting positions by attention mass; this is the direct upstream inspiration for the readout half of AGRS. None of these methods target sparse MoE routing.

(ii) Sparse Mixture-of-Experts language models. *Mixtral of Experts* (Jiang et al., 2024) and *OLMoE: Open Mixture-of-Experts Language Models* (Muennighoff et al., 2024) introduced the open-weight checkpoints used in this thesis. Each sparse layer routes every token to two experts via a learned router; this discrete selection is the structure behavioral steering exploits.

(iii) Steering sparse MoE models. SteerMoE (2024) is the upstream work that first proposed biasing router logits to alter which experts are selected at decode time. Its readout uses uniform averaging over a target span. The diagnostic results in my preliminary draft indicate that this readout produces a diffuse risk-difference signal on Mixtral and correspondingly small behavioral shifts under a question-only test. This thesis builds directly on SteerMoE's intervention site while replacing its readout.

(iv) Feature learning and AGOP. Radhakrishnan et al. (*Science*, 2024) introduce the *neural feature ansatz* and the *Average Gradient Outer Product (AGOP)* as a mechanism through which neural networks acquire features. *Mechanism of Feature Learning in Convolutional Neural Networks* (arXiv:2309.00570) extends AGOP to convolutional architectures. *Linear Recursive Feature Machines Provably Recover Low-rank Matrices* (arXiv:2401.04553) gives RFM-based algorithms with provable recovery guarantees. *Quadratic Models for Understanding Neural Network Dynamics* (ICLR 2024) uses quadratic surrogates to analyze training dynamics. The thesis uses AGOP to characterize which input-token directions a Mixtral expert is sensitive to, and asks whether AGRS's selections align with those directions.

8. Methodology

The project is mainly computational, in the sense IACS requires: every claim is empirical and reproducible from a pinned codebase, a pinned environment, and a fixed set of model checkpoints.

8.1 Models and compute

- *Models.* Mixtral 8x7B-Instruct-v0.1 (8-bit with CPU offload), OLMoE 1B-7B-0125-Instruct, Qwen2-MoE 14B-A2.7B-Instruct, and DeepSeek-V2-MoE-Lite-Chat if licensing and storage allow. Each is loaded once via Hugging Face Transformers under a recorded revision SHA.

- *Compute.* Two NVIDIA A100 GPUs per generation job on the MIT ORCD `mit_normal_gpu` partition. Total budget estimated at ~ 305 GPU-hours across the full plan, sized for a single graduate-student semester.

- *Software.* Python 3.11; PyTorch; Hugging Face Transformers; `bitsandbytes`; `accelerate`; `numpy`, `pandas`, `matplotlib` for analysis; Apptainer for cluster-portable images.

8.2 Computational pipeline

1. *Routing-trace collection.* For each contrastive prompt pair (p^+, p^-), capture per-token, per-layer router logits and attention weights via Hugging Face forward hooks. Slice to the shared statement-body target span.

2. *Risk-difference table.* Per concept and per (layer, expert) cell, compute the difference between the prefix-conditioned and unprefixed activation rates, with the AGRS variant weighting each token by its attention mass to the concept span at a chosen reference layer.

3. *Sparse plan selection.* Globally select the top k_+ positive cells and top k_- negative cells, subject to $|\Delta_{\ell,e}| > \varepsilon$.

4. *Decode-time intervention.* Add $\pm\alpha$ to the selected experts' router logits at every forward pass during generation.

5. *Evaluation.* Score each generation with an LLM judge (GPT-4o by default, Claude Opus 4.6 as cross-check) on a 0–2 ordinal scale for thematic concept transfer. Cross-validate against a 200-case human-rater subset (Cohen's κ, Spearman). Report paired bootstrap 95% CIs over seeds.

8.3 Mechanistic analysis with AGOP

For a selected expert e at layer ℓ, treat the expert's gating gradient with respect to the input embedding sequence as the sensitivity direction. Estimate the AGOP

$$M_{\ell,e} \;=\; \mathbb{E}_x \left[\nabla_x f_{\ell,e}(x)\, \nabla_x f_{\ell,e}(x)^\top \right]$$

over the contrastive prompt pool, and compare its top eigenvectors against the concept-prototype embedding direction. The thesis question this analysis answers: *do the experts AGRS selects respond to feature directions that an AGOP-style analysis would identify as concept-relevant, or does AGRS find them incidentally through the activation-rate differential alone?*

8.4 Reproducibility commitments

- All generation calls at temperature 0, top-p 1.0, with explicit seeds.

- Every model load asserts the recorded HF revision SHA.

- Every CSV output row carries the git SHA of the codebase, the cluster job ID, and a wallclock stamp.
- One Apptainer image is the only supported runtime.
- Pre-registration of the AGRS hypothesis on OSF before any AGRS-vs-SteerMoE comparison is run.
- Anonymized public release of the full `experiments/` artifact tree under CC-BY at submission, so reviewers can re-derive every figure offline.

8.5 Phasing and risk

The eight-phase rollout (reproducibility hardening, evaluation harness, AGRS implementation, pilot, full method comparison, cross-model, cross-concept, mechanistic analysis + writing) is documented in the project's `paper/docs/REVISION_PLAN.md`, with compute budgets, exit criteria, and an explicit Phase-3 decision gate. The largest methodological risk is that AGRS does not clear the pre-registered $\Delta\mathrm{CTR} \geq +0.10$ threshold on the Mixtral / fears pilot; in that case the thesis pivots to either a same-readout-different-site comparison or to a more focused negative result, both of which preserve the AGOP analysis as the mechanistic core.

Companion materials in the project repository:

- Working conference draft: `paper/docs/paper/main.pdf`.
- Phased revision plan: `paper/docs/REVISION_PLAN.md`.
- Existing routing and steering backends: `src/moe_attention_guided_steering/`.